

When should we use "matrix inverse" instead of "Gaussian elimination" (GE), and vice versa, to solve linear system?

Answer:

While both methods can solve a linear system, they are designed for different situations and have significant differences in applicability and efficiency.

The short answer is: For solving a single linear system, Gaussian elimination is almost always the better, faster, and more versatile method. The matrix inverse method is primarily useful for theoretical purposes or in specific situations where you need to solve the same system with many different constant terms.

Gaussian Elimination (GE)

- **How it works:** You take the augmented matrix $[A \mid b]$ and use row operations to transform it into Row Echelon Form (REF). Then, you use back substitution to find the solution.
- **Applicability:** Universal. It works for any system of linear equations:
 - Square systems (n equations, n variables) and non-square systems (more or fewer equations than variables).
 - Systems with a unique solution, no solution, or infinitely many solutions. GE will clearly reveal which case you are dealing with.

Matrix Inverse Method

- **How it works:** For a system $Ax = b$, the solution is given by the formula $x = A^{-1}b$. This requires you to first find the inverse of the coefficient matrix A , and then multiply it by the vector b .
- **Applicability:** Highly restricted. It only works if:
 1. The system is square (the number of equations equals the number of variables).
 2. The coefficient matrix A is non-singular ($\det(A) \neq 0$). This means the system must have a unique solution.

This method cannot be used if the system is non-square, has no solution, or has infinitely many solutions.

For theoretical work, proofs, or deriving formulas, matrix inverse has the advantages. The notation $x = A^{-1}b$ is incredibly compact and powerful for algebraic manipulation. In fields like control theory, statistics, computer graphics, and robotics, the inverse is a fundamental concept used to derive general formulas and understand the properties of a system, even if it's not the method used for the final computation.

Scenario 1: Solving linear system where coefficient matrix is non-singular

$$\begin{cases} x + 2y + z = 7 \\ y + 2z = 9 \\ x + 3z = 8 \end{cases}$$

This system can be written in matrix equation form $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 3 \end{pmatrix}$ and

$$\mathbf{b} = \begin{pmatrix} 7 \\ 9 \\ 8 \end{pmatrix}.$$

1. Solution using Gaussian Elimination (GE)

The goal of Gaussian Elimination is to transform the augmented matrix $[A \mid b]$ into row-echelon form and then use back-substitution to find the variables.

Our augmented matrix is: $\begin{bmatrix} 1 & 2 & 1 & 7 \\ 0 & 1 & 2 & 9 \\ 1 & 0 & 3 & 8 \end{bmatrix}$.

After performing GE, we obtain the row echelon form $\begin{bmatrix} 1 & 2 & 1 & 7 \\ 0 & 1 & 2 & 9 \\ 0 & 0 & 6 & 19 \end{bmatrix}$

Use back-substitution to solve for x , y , and z .

From Row 3: $6z = 19 \Rightarrow z = \frac{19}{6}$

From Row 2: $y + 2z = 9 \Rightarrow y = 9 - 2z = 9 - 2\left(\frac{19}{6}\right) = 9 - \frac{19}{3} = \frac{27-19}{3} = \frac{8}{3}$

From Row 1: $x + 2y + z = 7 \Rightarrow x = 7 - 2y - z = 7 - 2\left(\frac{8}{3}\right) - \frac{19}{6} = 7 - \frac{16}{3} - \frac{19}{6} = \frac{42}{6} - \frac{32}{6} - \frac{19}{6} = \frac{42-32-19}{6} = \frac{10-19}{6} = -\frac{9}{6} = -\frac{3}{2}$

The solution is $x = -3/2$, $y = 8/3$, $z = 19/6$.

2. Solution using the Matrix Inverse Method

Since the matrix $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 3 \end{pmatrix}$ is non-singular, we can find its inverse $\mathbf{A}^{-1}$, and the

solution to the system $\mathbf{Ax} = \mathbf{b}$ is given by $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

We first find the inverse of $\mathbf{A}$. either using GJE or MATLAB:

$$\mathbf{A}^{-1} = \frac{1}{6} \begin{pmatrix} 3 & -6 & 3 \\ 2 & 2 & -2 \\ -1 & 2 & 1 \end{pmatrix}$$

Calculate the solution vector $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 3 & -6 & 3 \\ 2 & 2 & -2 \\ -1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 7 \\ 9 \\ 8 \end{pmatrix} = \begin{pmatrix} -\frac{3}{2} \\ \frac{8}{3} \\ \frac{19}{6} \end{pmatrix}$$

The solution is $x = -3/2$, $y = 8/3$, $z = 19/6$.

Both methods yield the same unique solution, as expected for a system with a non-singular coefficient matrix.

Scenario 2: Solving linear system where coefficient matrix is singular

$$\begin{cases} x + 2y + 3z = 4 \\ y + 2z = 0 \\ x + 3y + 5z = 5 \end{cases}$$

This system can be written in matrix equation form $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 3 & 5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ 1 \\ 5 \end{pmatrix}$.

We check that the determinant of $\mathbf{A}$ is 0, and hence the matrix is singular. Therefore its inverse $\mathbf{A}^{-1}$ does not exist, and the matrix inverse method does not work.

The only way to solve such a system is with a method like **Gaussian Elimination (GE)**.

Reduce the augmented matrix of the system $[\mathbf{A} \mid \mathbf{b}]$ to a row echelon form:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & 3 & 5 & 5 \end{bmatrix} \xrightarrow{GE} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

We have fewer independent equations than variables, leading to infinite solutions.

To describe these solutions, we introduce a parameter. Let $z = t$, where t is any real number.

From Row 2: $y + 2z = 1 \Rightarrow y = 1 - 2z = 1 - 2t$

From Row 1: $x + 2y + 3z = 4 \Rightarrow x = 4 - 2y - 3z = 4 - 2(1 - 2t) - 3(t) = 4 - 2 + 4t - 3t = 2 + t$

Conclusion: The system has infinitely many solutions, with general solution:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 + t \\ 1 - 2t \\ t \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \text{ for any real number } t.$$

Scenario 3: Solving linear system where coefficient matrix is non-square

$$\begin{cases} x + 2y + z + 3w = 6 \\ 2x + 5y + z + 4w = 10 \\ x + y + 2z + 5w = 8 \end{cases}$$

This system can be written in matrix equation form $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 5 & 1 & 4 \\ 1 & 1 & 2 & 5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 6 \\ 10 \\ 8 \end{pmatrix}$.

Important Note on Methods:

- **Matrix Inverse Method:** This method is **not applicable** for a 3×4 system. The inverse of a matrix is only defined for square matrices ($n \times n$ matrices). Since a 3×4 matrix is not square, it cannot have an inverse.

- **Gaussian Elimination (GE):** This is the universal method for solving any system of linear equations, regardless of its dimensions. It is the correct method to use here.

Solving with Gaussian Elimination

We will transform the augmented matrix $[\mathbf{A} \mid \mathbf{b}]$ into row-echelon form.

$$\begin{bmatrix} 1 & 2 & 1 & 3 & 6 \\ 2 & 5 & 1 & 4 & 10 \\ 1 & 1 & 2 & 5 & 8 \end{bmatrix} \xrightarrow{GE} \begin{bmatrix} 1 & 2 & 1 & 3 & 6 \\ 0 & 1 & -1 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The non-pivot columns are columns 3 and 4. These correspond to the variables z and w , which we assign parameters $z = s, w = t$.

Now, use back-substitution to solve for the pivot variables in terms of s and t .

$$\text{From Row 2: } y - z - 2w = -2 \Rightarrow y = -2 + z + 2w = -2 + s + 2t$$

$$\text{From Row 1: } x + 2y + z + 3w = 6 \Rightarrow x = 6 - 2y - z - 3w$$

Substitute the expressions for y, z , and w :

$$x = 6 - 2(-2 + s + 2t) - s - 3t = 6 + 4 - 2s - 4t - s - 3t = 10 - 3s - 7t$$

The general solution is:

- $x = 10 - 3s - 7t$
- $y = -2 + s + 2t$
- $z = s$
- $w = t$

Scenario 4: Solve multiple square systems with same coefficient matrix (non-singular) but different constant terms

The core idea is that the most computationally intensive part of solving a linear system is the work done on the coefficient matrix $\mathbf{A}$. If $\mathbf{A}$ remains the same, we can perform this work just once and reuse it to solve for any number of different constant vectors $\mathbf{b}$.

Let's use a non-singular 3x3 coefficient matrix $\mathbf{A}$ and two different constant vectors, $\mathbf{b}_1$ and $\mathbf{b}_2$.

- Coefficient Matrix: $\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{pmatrix}$ (We can check that it is non-singular.)
- Constant Vectors: $\mathbf{b}_1 = \begin{pmatrix} 9 \\ 16 \\ 9 \end{pmatrix}$ and $\mathbf{b}_2 = \begin{pmatrix} 4 \\ 7 \\ 4 \end{pmatrix}$

We need to solve two separate systems:

1. System 1: $\mathbf{Ax} = \mathbf{b}_1$
2. System 2: $\mathbf{Ay} = \mathbf{b}_2$ (using $\mathbf{y}$ for the second solution vector to avoid confusion)

1. Solution using Gaussian Elimination (Augmented Matrix)

Instead of solving each system separately, we can augment the matrix $\mathbf{A}$ with both constant vectors at the same time. This allows us to perform the row operations on $\mathbf{A}$ just once.

Our combined augmented matrix is $[\mathbf{A} \mid \mathbf{b}_1 \mid \mathbf{b}_2]$:
$$\begin{bmatrix} 1 & 1 & 2 & 9 & 4 \\ 2 & 3 & 2 & 16 & 7 \\ 1 & 2 & 1 & 9 & 4 \end{bmatrix}$$

Now, we apply Gaussian Elimination to transform the left side ($\mathbf{A}$) into row-echelon form:

$$\begin{bmatrix} 1 & 1 & 2 & 9 & 4 \\ 0 & 1 & -2 & -2 & -1 \\ 0 & 0 & 1 & 2 & 1 \end{bmatrix}$$

The matrix is now in row-echelon form with the last two columns the resulting “constant vectors” after the reduction. We can now solve each system by back-substitution.

Solving System 1 (using the first constant column): The system is:

- $x_1 + x_2 + 2x_3 = 9$
- $x_2 - 2x_3 = -2$
- $x_3 = 2$

From Row 3: $x_3 = 2$

From Row 2: $x_2 = -2 + 2x_3 = -2 + 2(2) = 2$

From Row 1: $x_1 = 9 - x_2 - 2x_3 = 9 - 2 - 2(2) = 3$

Solution 1: $\mathbf{x} = (3, 2, 2)$

Solving System 2 (using the second constant column): The system is:

- $y_1 + y_2 + 2y_3 = 4$
- $y_2 - 2y_3 = -1$
- $y_3 = 1$

From Row 3: $y_3 = 1$

From Row 2: $y_2 = -1 + 2y_3 = -1 + 2(1) = 1$

From Row 1: $y_1 = 4 - y_2 - 2y_3 = 4 - 1 - 2(1) = 1$

Solution 2: $\mathbf{y} = (1, 1, 1)$

2. Solution using the Matrix Inverse Method

This method shines in this scenario. The strategy is to compute the inverse of $\mathbf{A}$, $\mathbf{A}^{-1}$, only once. Then, we can find each solution by a simple matrix-vector multiplication.

- $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}_1$
- $\mathbf{y} = \mathbf{A}^{-1}\mathbf{b}_2$

We find $\mathbf{A}^{-1}$ using GJE or MATLAB:
$$\mathbf{A}^{-1} = \begin{pmatrix} -1 & 3 & -4 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{pmatrix}$$

Solving System 1: $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}_1 = \begin{pmatrix} -1 & 3 & -4 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 9 \\ 16 \\ 9 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$

Solution 1: $\mathbf{x} = (3, 2, 2)$

Solving System 2: $\mathbf{y} = \mathbf{A}^{-1}\mathbf{b}_2 = \begin{pmatrix} -1 & 3 & -4 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 7 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

Solution 2: $\mathbf{y} = (1, 1, 1)$

Comparison and Conclusion

Both methods yield the exact same solutions, as expected.

- The **Augmented Gaussian Elimination** method is very efficient because all row operations are done simultaneously. However, the back substitution has to be done individually for each set of constant terms. It is a robust and direct way to find the solutions, and it works regardless of whether the coefficient matrix is square or no-square, singular or non-singular.
- The **Matrix Inverse** method is conceptually very powerful. Once you have $\mathbf{A}^{-1}$, solving for any new vector $\mathbf{b}$ is just a matter of a quick multiplication. This method is especially advantageous if you need to solve the system for many different constant vectors, as the expensive computation (finding the inverse) is done only once.